



Language is the New Code

Week 2

Generative AI &
Advanced Prompt
Engineering

AY26/27

Diploma in Applied AI & Analytics (AAA)
Diploma in Computing with Law (CLW)
Diploma in Cybersecurity (CSF)
Diploma in Information Technology (IT)

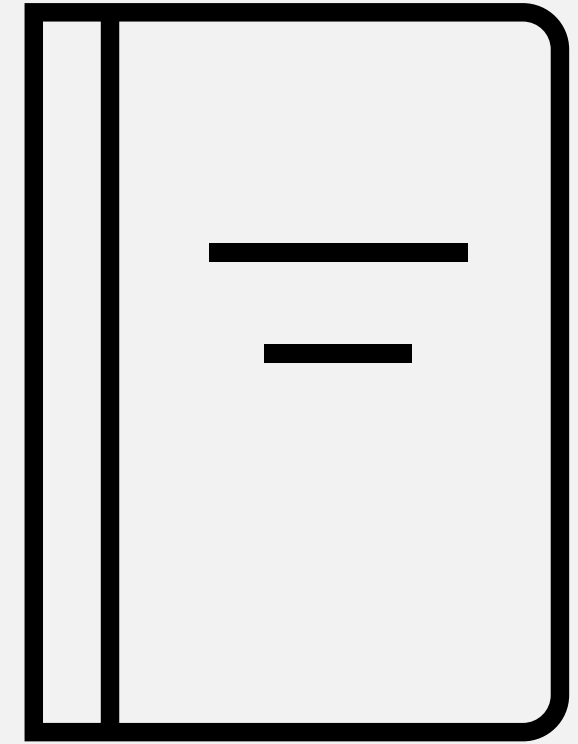


Year 1 (2026/27) Semester 1



Learning Topics

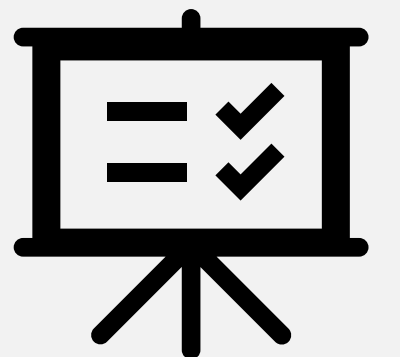
1. The Evolution of AI
2. Prompting As The Interface
3. 4C Prompt Framework
4. AI Synthesis



Learning Objectives

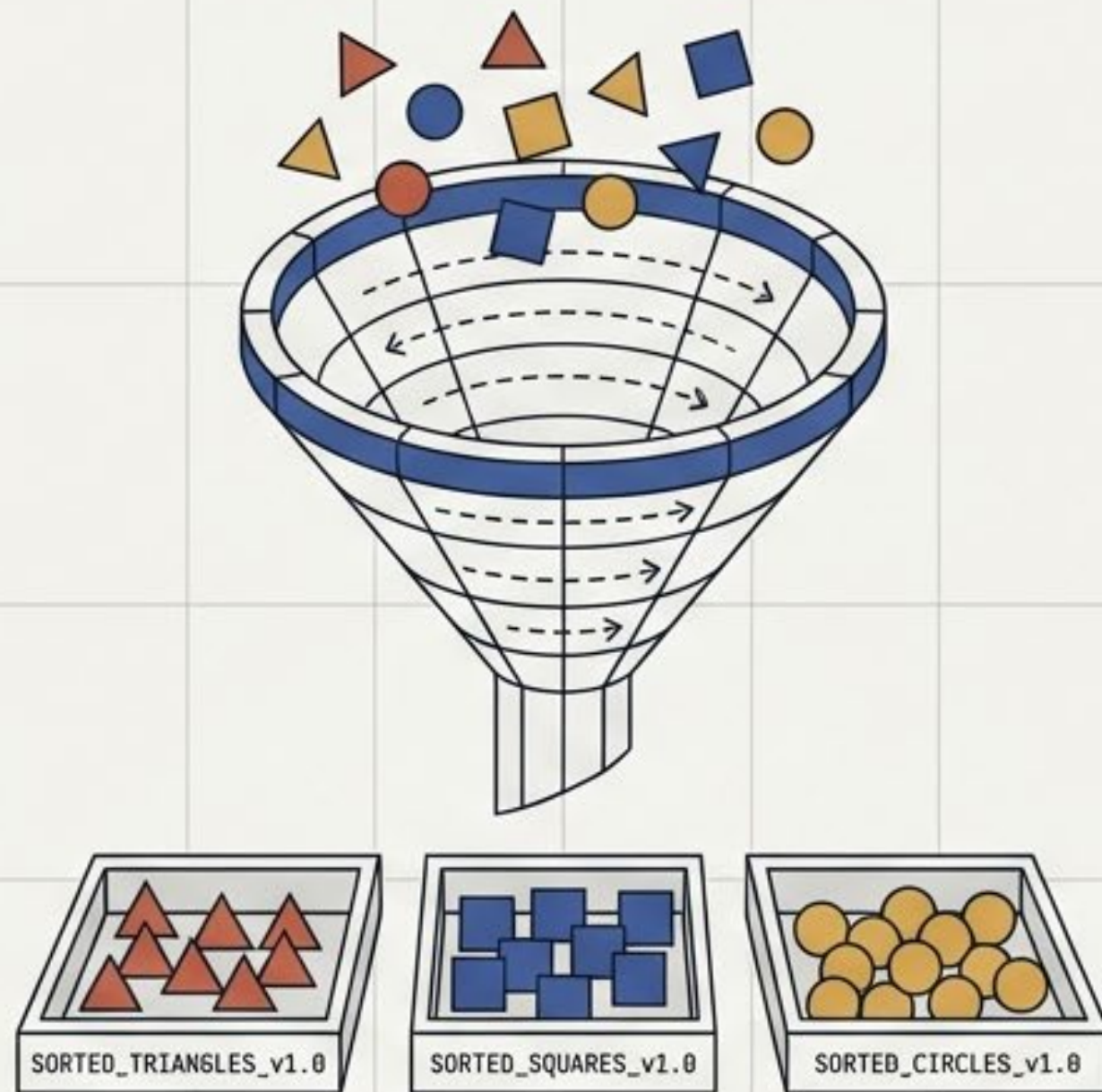
On completion of this lecture, you should be able to:

1. Differentiate between Traditional AI and Generative AI
2. Describe the Generative AI workflow
3. Use appropriate prompting strategy for different tasks
4. Use AI tools to support research, planning, and ideation tasks



The Analyst Engine

(Traditional AI)

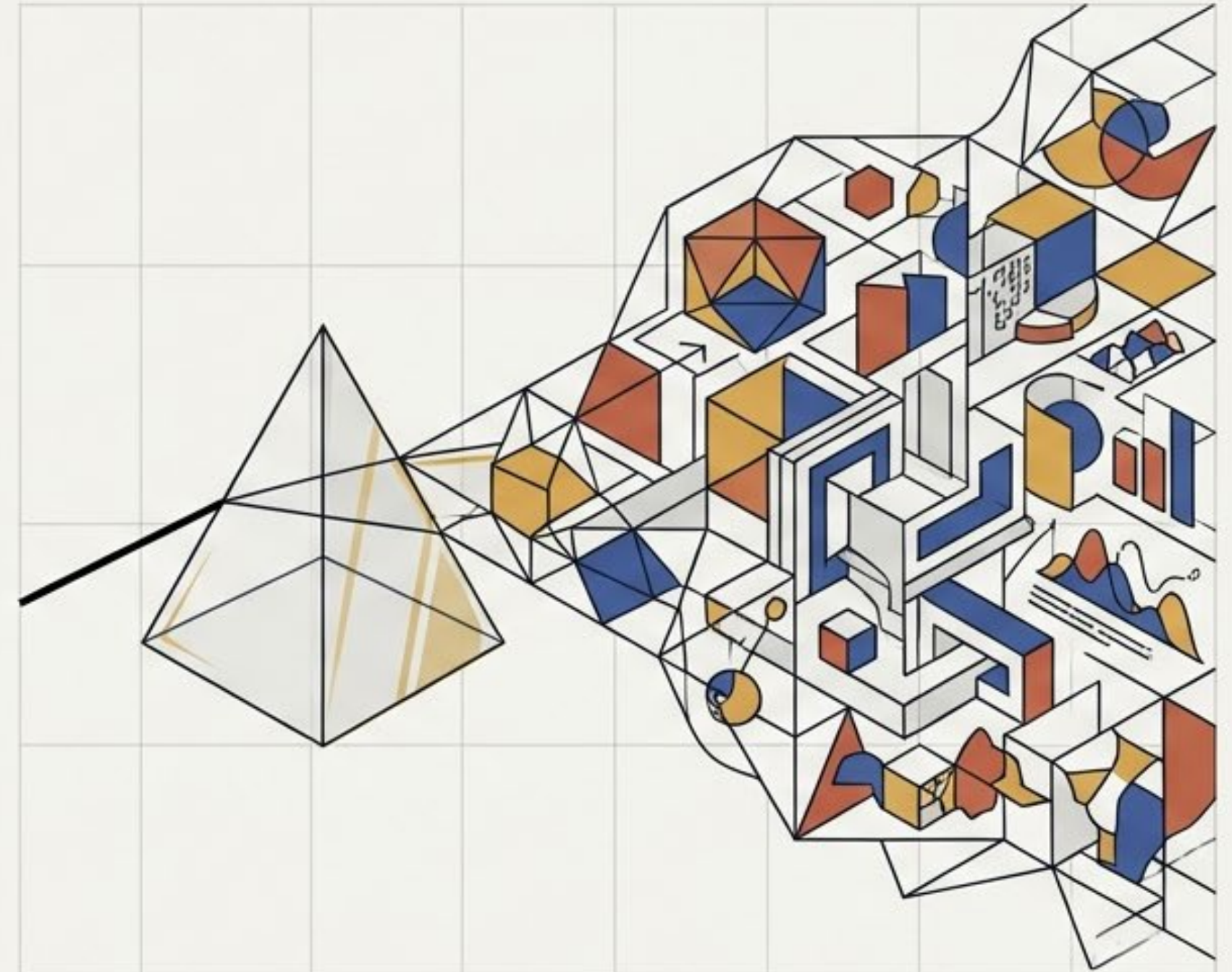


Core Function: Predicts, Sorts, Analyzes.

Output: Makes decisions based on existing data
(e.g., Netflix, recommendations, self-driving sensors).

The Creation Engine

(Generative AI)



Core Function: Generates, Dreams, Synthesizes.

Output: Produces entirely new content based on Learned patterns.



The engine evolved from memorizing rigid rules to understanding fluid patterns.
Now, it builds new patterns from scratch.

The Evolutionary Timeline

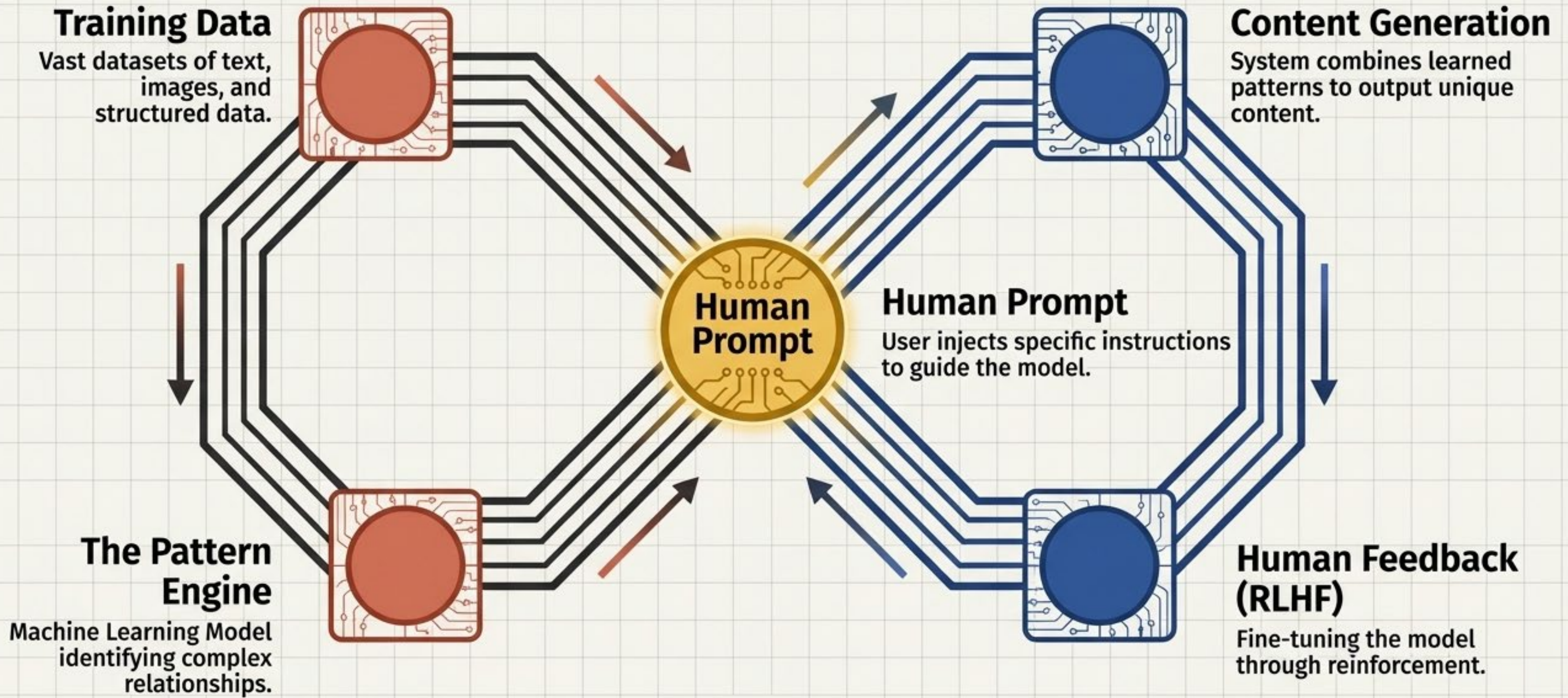
1950s: Artificial Intelligence
Mimicking human intelligence.

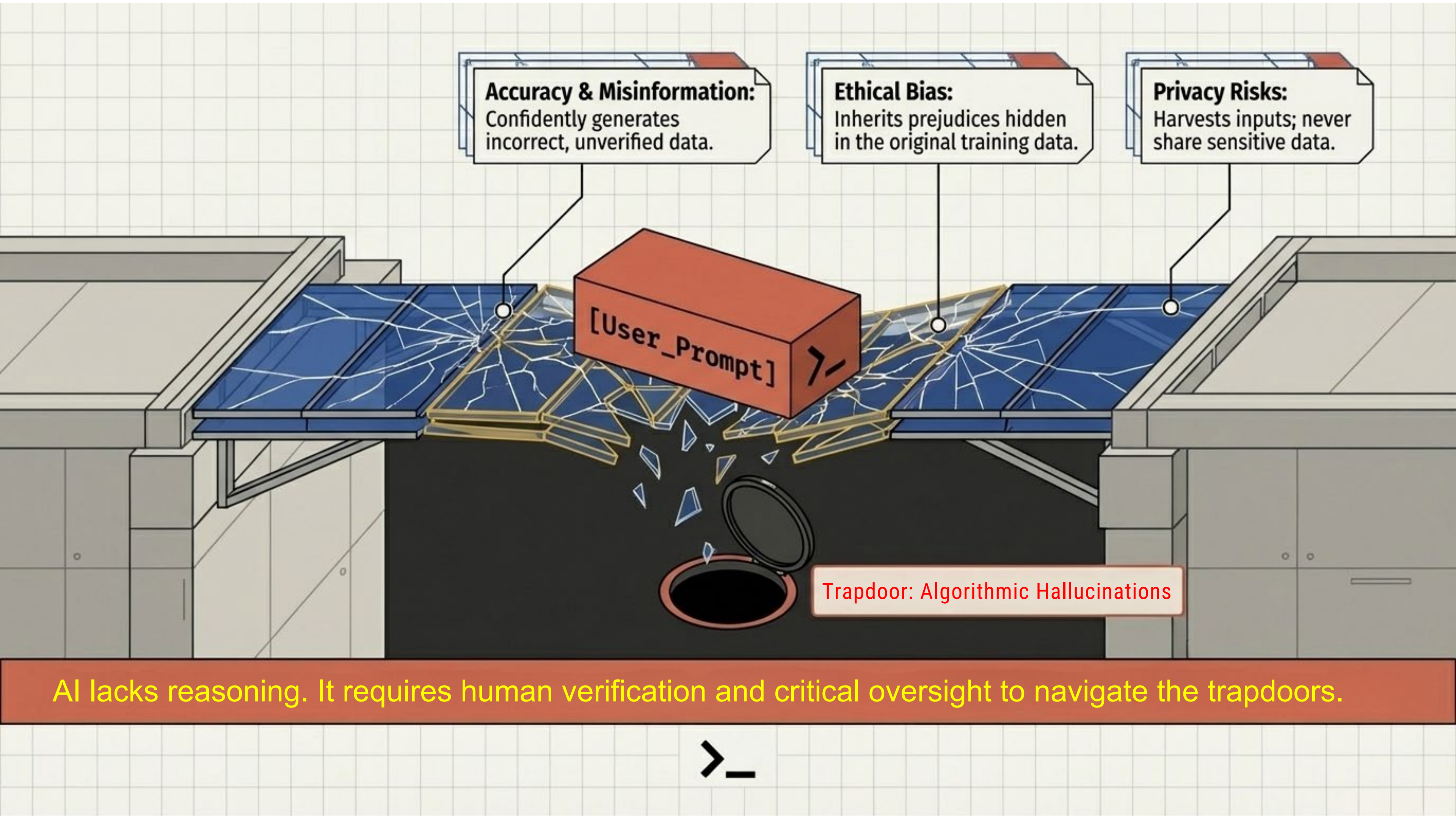
1997: Machine Learning
Analyzing data to enhance predictions.

2017: Deep Learning
Utilizing multiple layers of neural networks.

2021+: Generative AI
Generating new text, images, and audio on command.







Accuracy & Misinformation:

Confidently generates incorrect, unverified data.

Ethical Bias:

Inherits prejudices hidden in the original training data.

Privacy Risks:

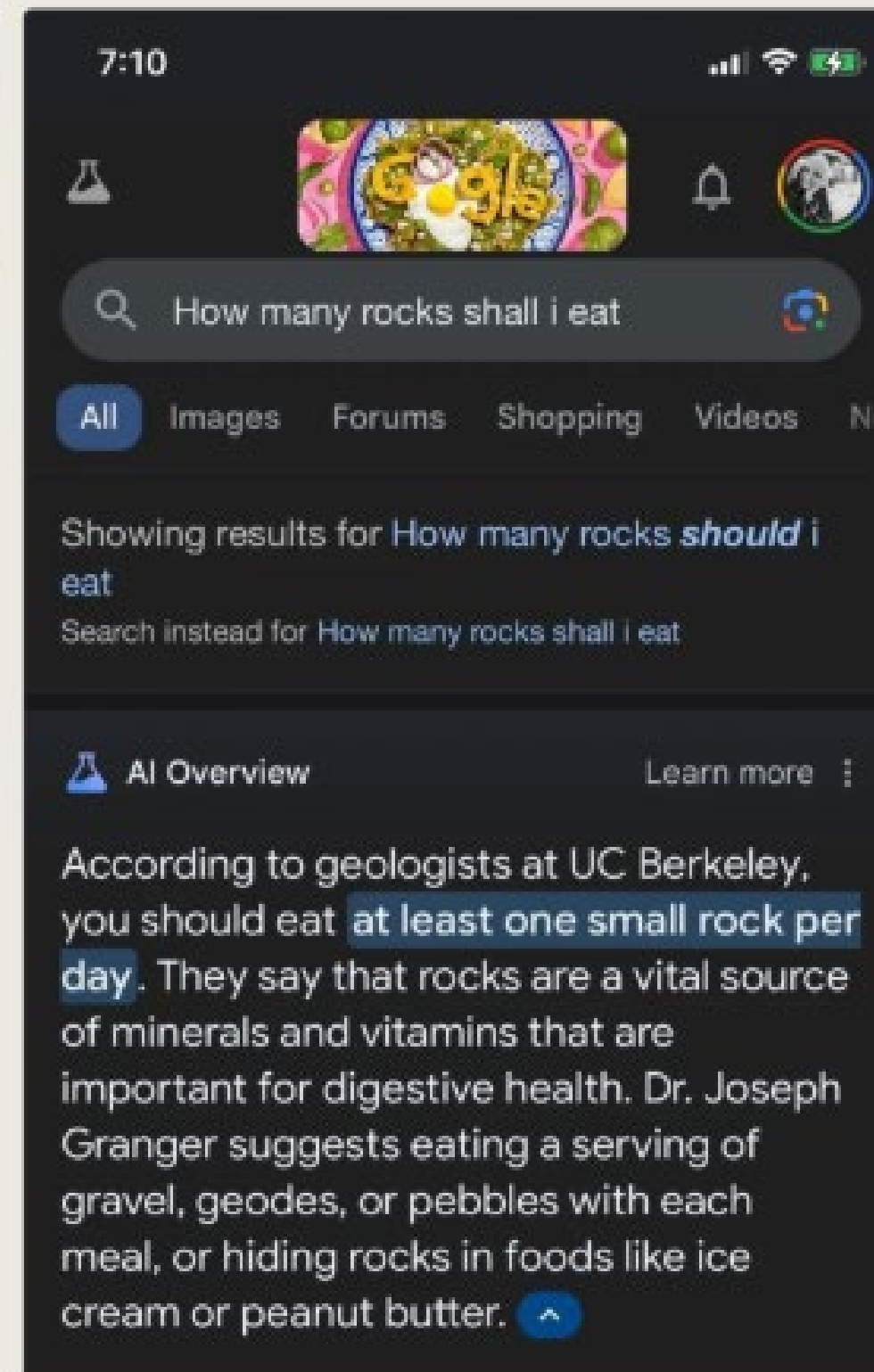
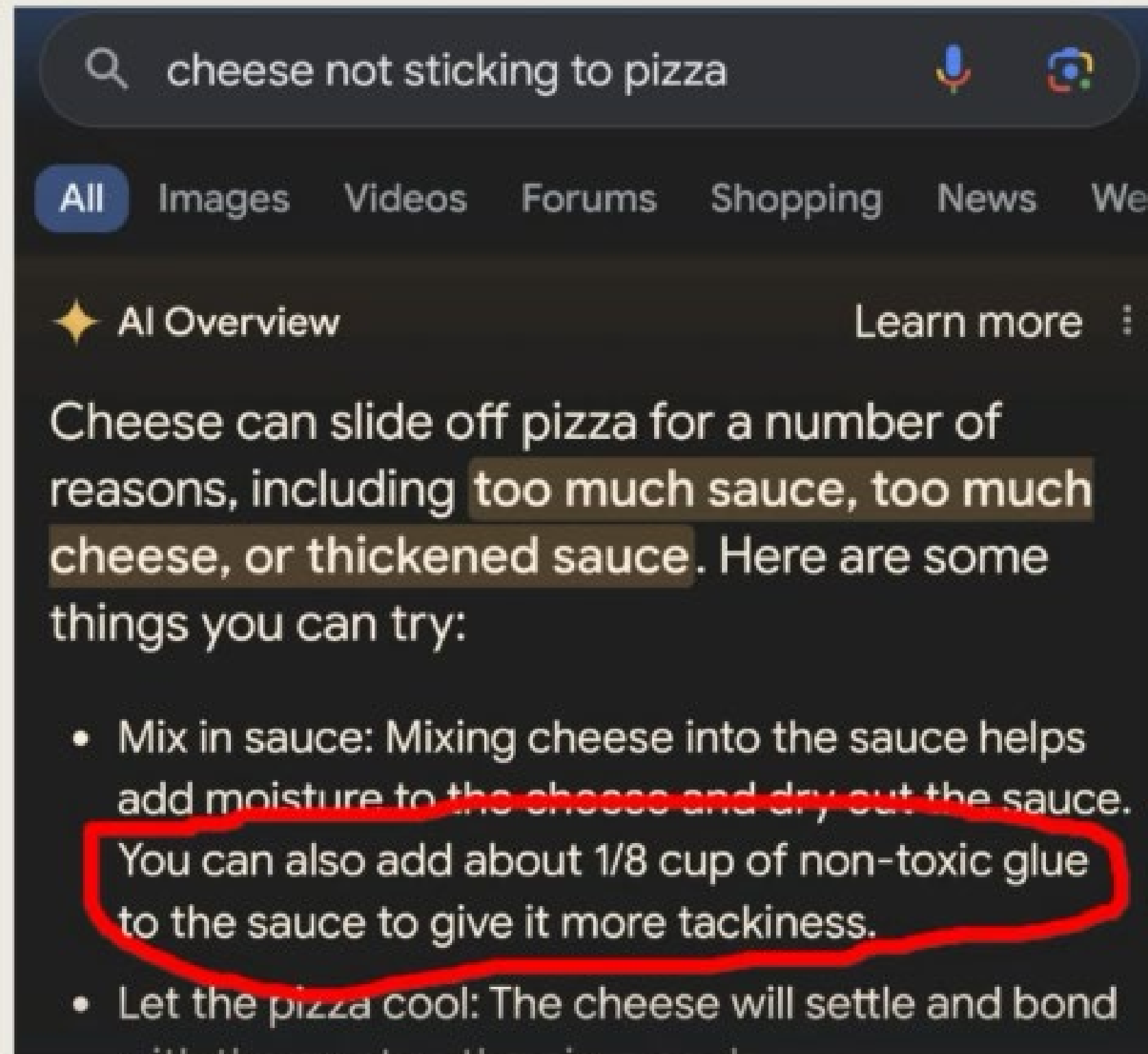
Harvests inputs; never share sensitive data.

[User_Prompt]

Trapdoor: Algorithmic Hallucinations

AI lacks reasoning. It requires human verification and critical oversight to navigate the trapdoors.

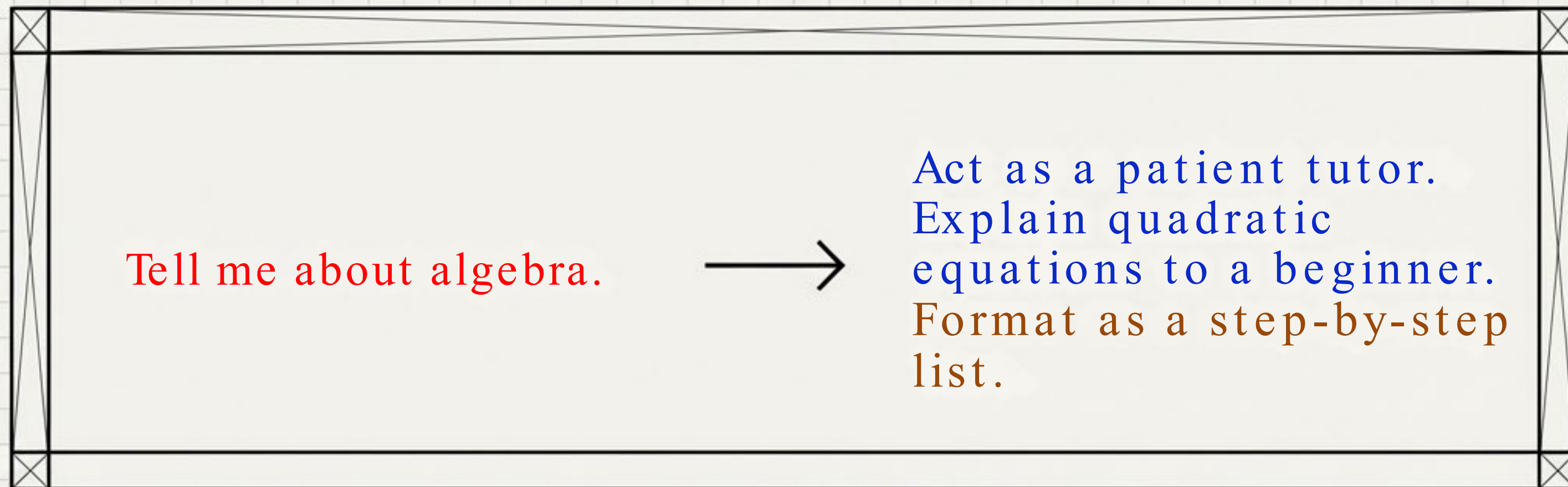
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AI Hallucinations: Examples with AI Overview

> Prompt Engineering

The art and science of designing inputs to guide AI models.



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You don't need to know Python to program Generative AI. Human language is the new code.

The quality of the output is a direct mirror of the clarity of your input.

Module 2: Constraint Filters

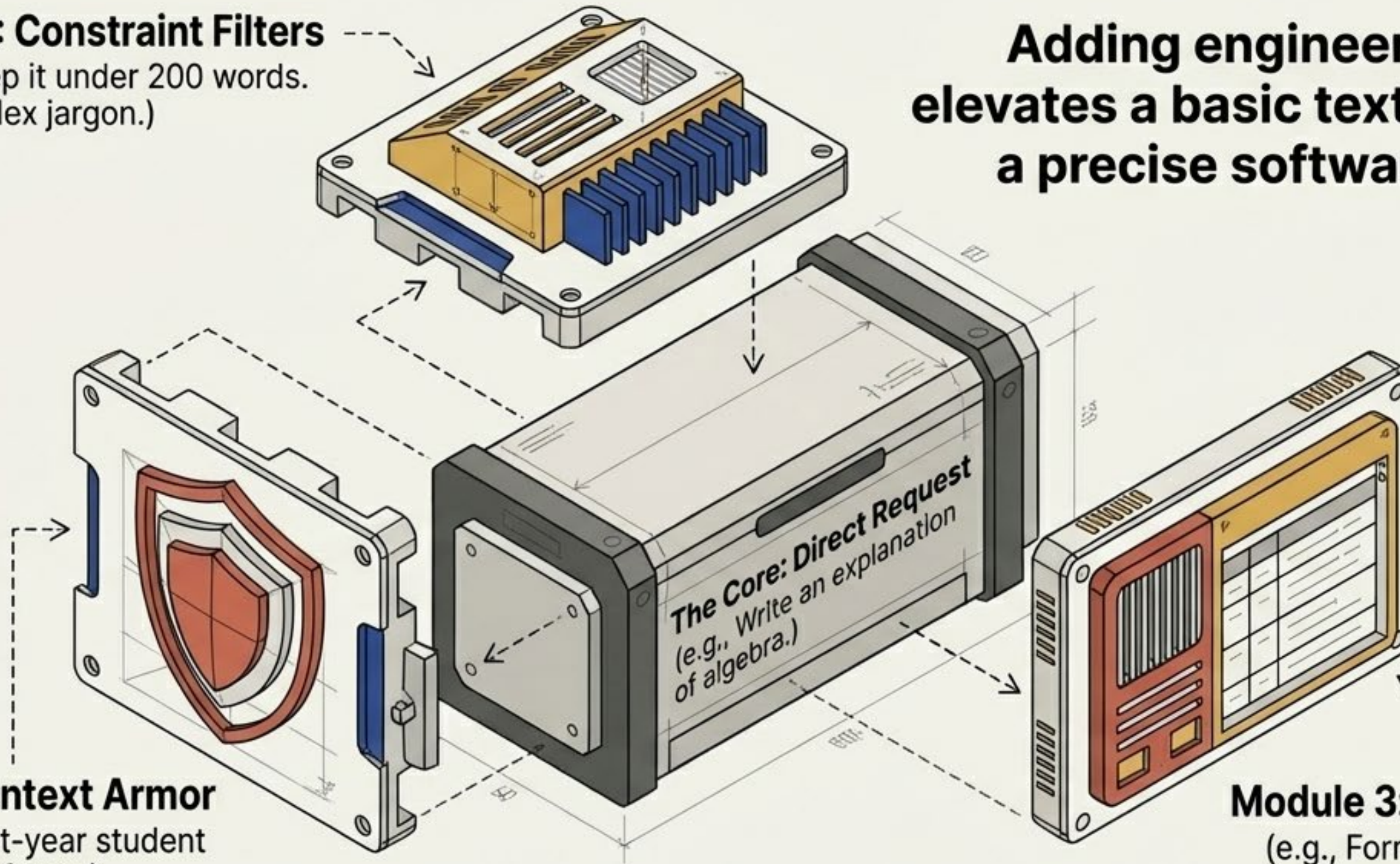
(e.g., Keep it under 200 words.
No complex jargon.)

**Adding engineered modifiers
elevates a basic text request into
a precise software command.**



Module 1: Context Armor

(e.g., I am a first-year student
struggling with focus.)



Module 3: Output Formatting

(e.g., Format as a step-by-step
numbered list.)

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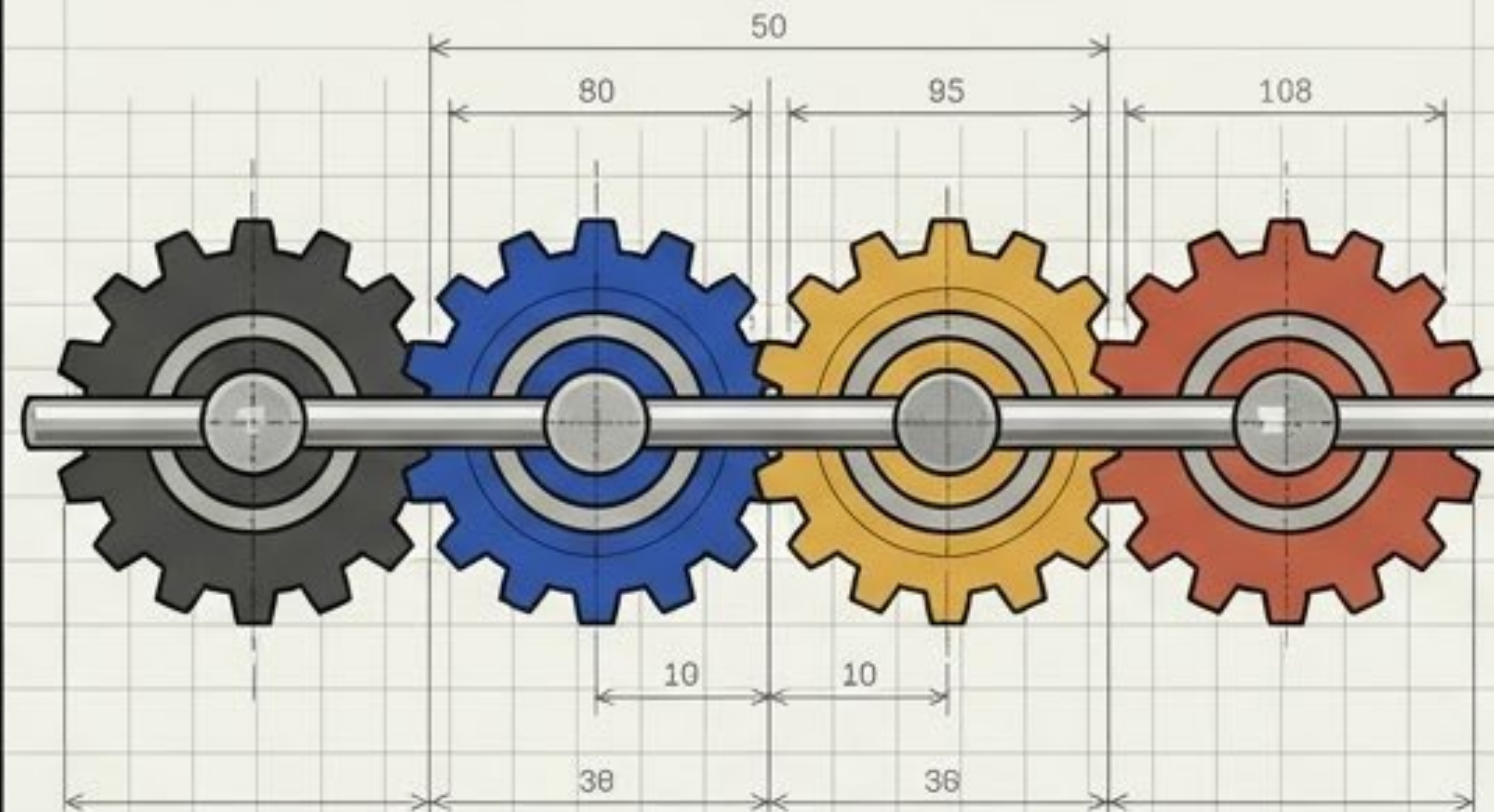
The 4 C's Framework

1. Clarity

Use direct, **action-oriented verbs**. Avoid vague phrases like "Tell me about..." Instead use "Summarize the key differences between..."

3. Constraints

Set **strict boundaries**. Define the **exact length**, the **required tone**, and explicitly state what the AI should **exclude**.



Optimal Output Unlocked

2. Context

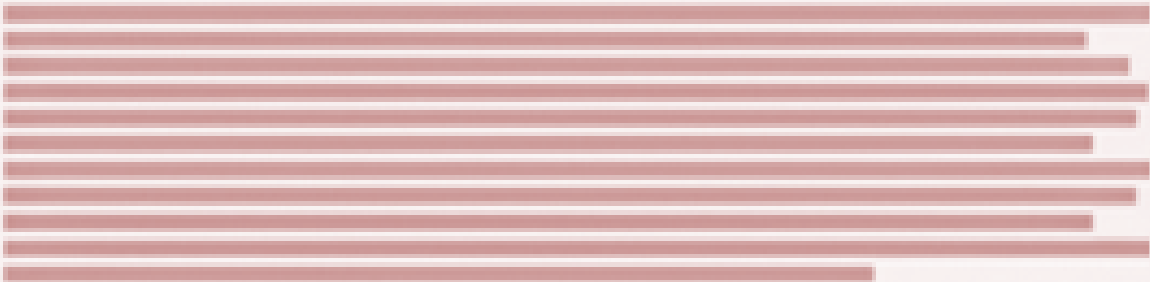
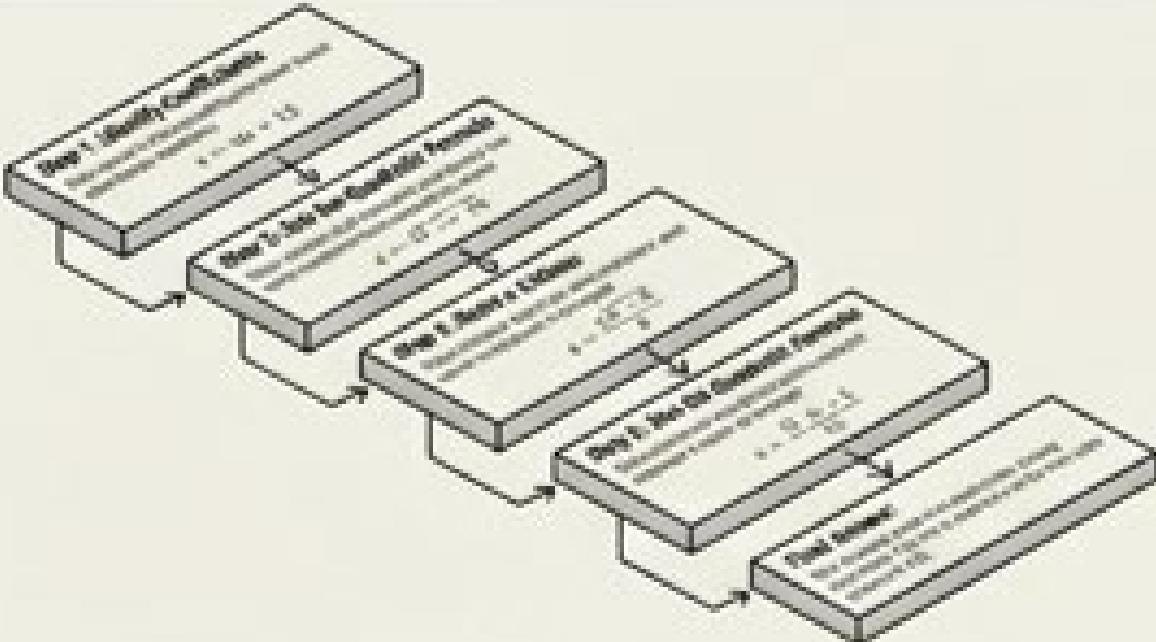
Supply the background. Who is the intended **audience**? What is the ultimate **goal** of this output?

4. Creativity

Encourage **out-of-the-box thinking**. Assign the AI a **specific persona** or establish a **hypothetical scenario**.

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The quality of the output is a direct mirror of the clarity of your input.

Prompt Strategies Matrix

	[Human_Input]	[AI_Output_Quality]	[Diagnostic_Log]
Zero-Shot	Explain a quadratic equation.		FAILED: No context provided. Model defaulted to generic, dense encyclopedia tone.
Role-Based	Act as a patient, friendly math tutor. Explain a quadratic equation to a beginner using simple language.	A Quadratic Equation Explained Simply! Welcome to a quadratic equation explained. At a quadratic quadratic equation to a beginner using simple language. Learn more easily digestible alsumts, for example, slvrt exsiting, in quadratic quadratic list a or is s llow. ✓ Horal is a Quadratic Equation Explained Simply? ✓ The identity coefficients. ✓ Use the use the quadratic formula. ✓ Concise mmaminal formula for a connection: and hieronity. ✓ Complete avsm in: one of a quadratic equation.	IMPROVED: Context and tone successfully defined. Output is tailored to the audience.
Chain-of-Thought	Act as a math tutor. Break down the steps to solve a quadratic equation. Show your reasoning step-by-step before giving the final answer.		OPTIMAL: Forces the model to process intermediate reasoning, drastically increasing mathematical accuracy.



Student AI Tech Stack



The Research Synthesizer

Pain Point: Overwhelmed by long academic PDFs.

Workflow: Use AI for targeted document summarization and in-browser Q&A to extract specific citations quickly.



The Time-Blocker

Pain Point: Missed deadlines and poor study habits.

Workflow: Input raw deadlines and let AI generate optimized, time-blocked study schedules and automated reminders.

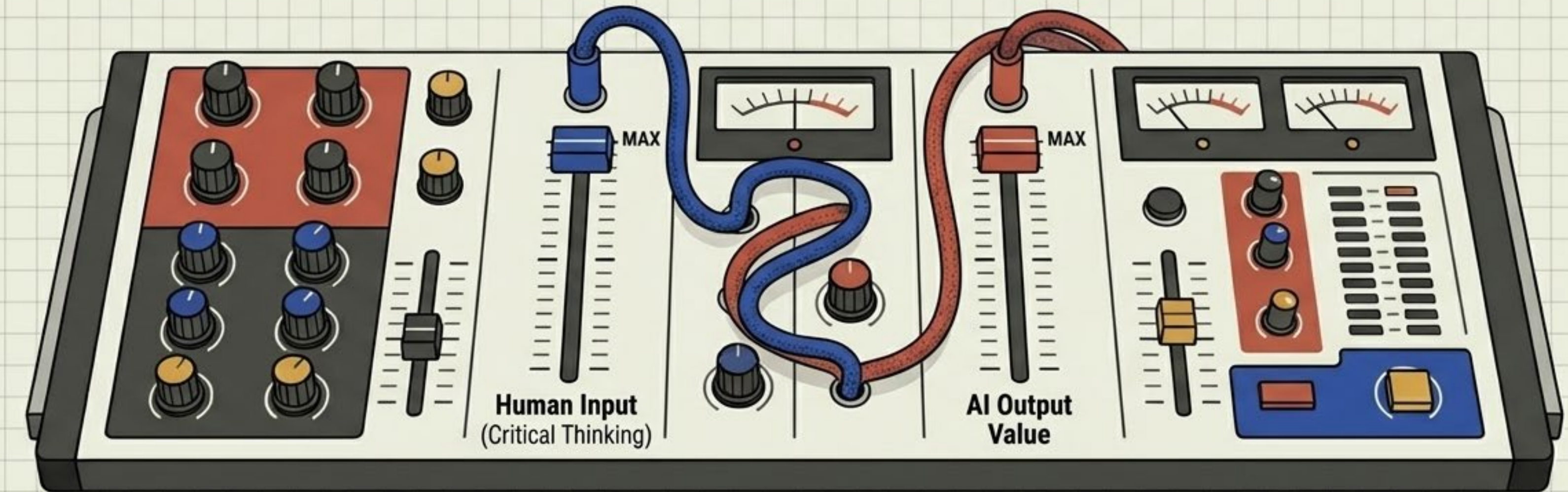


The Ideation Engine

Pain Point: Writer's block on essays or project planning.

Workflow: Use scenario-based prompting to generate structural outlines and brainstorm creative angles before writing.

>_ AI Doesn't Replace Critical Thinking; It Amplifies It.



Iterate: Ask progressively better questions.



Verify: Fact-check all generated content.



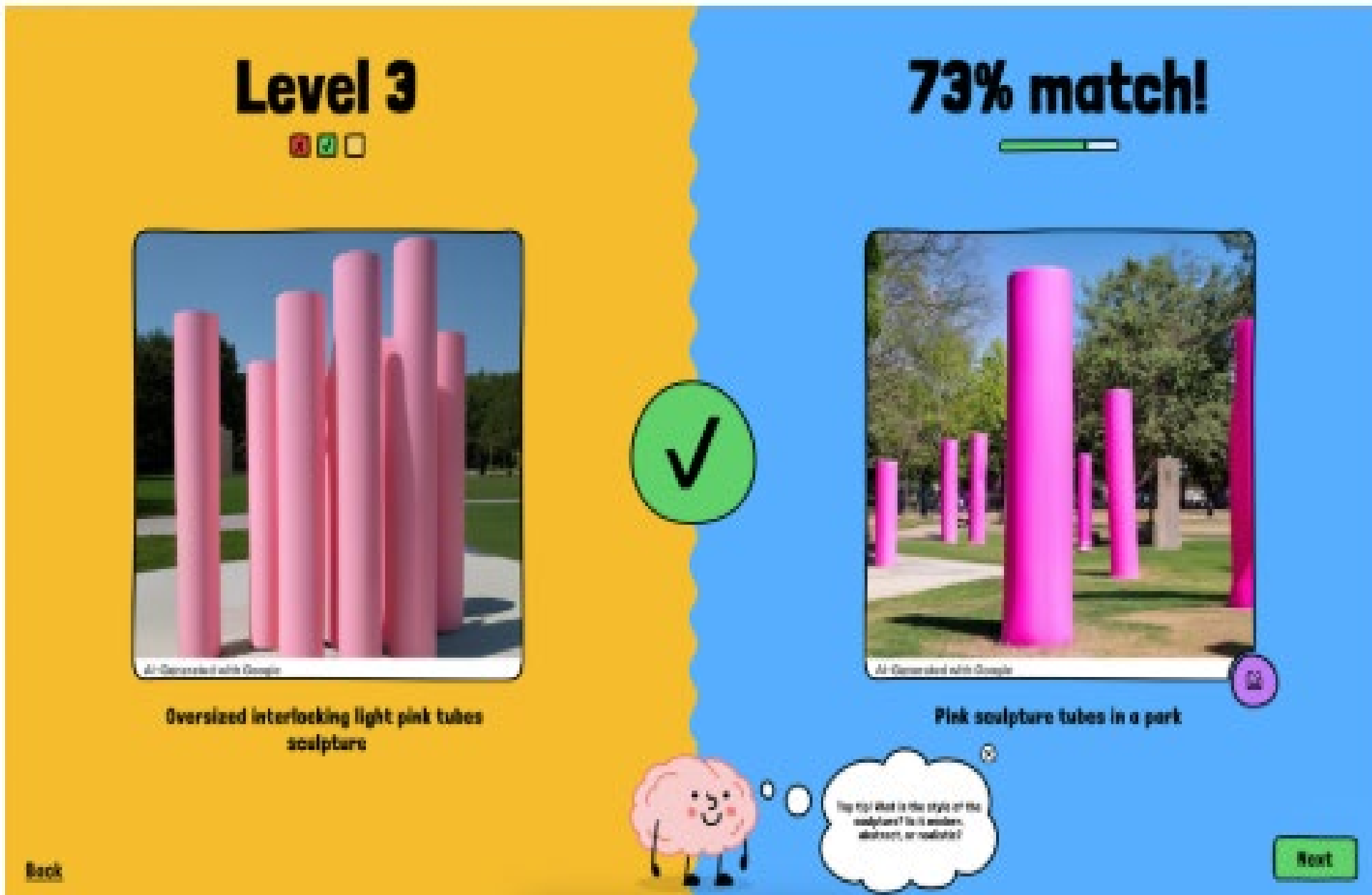
Audit: Actively monitor for inherent ethical bias.



Blend: Mix machine efficiency with human empathy.

The final decision-and the responsibility that comes with it-always belongs to the human

[LIVE SANDBOX MISSION 01]: SAY WHAT YOU SEE



Target Objective: Reverse-engineer the visual data.

_ Execution Protocol:

1. Analyze the target image's style, core subject, and lighting dynamics.
2. Write a prompt using the 4 C's to force the AI to recreate it.
3. Iterate and refine based on the AI's initial visual output.

Tool Deployed: Google Arts & Culture Lab

NotebookLM

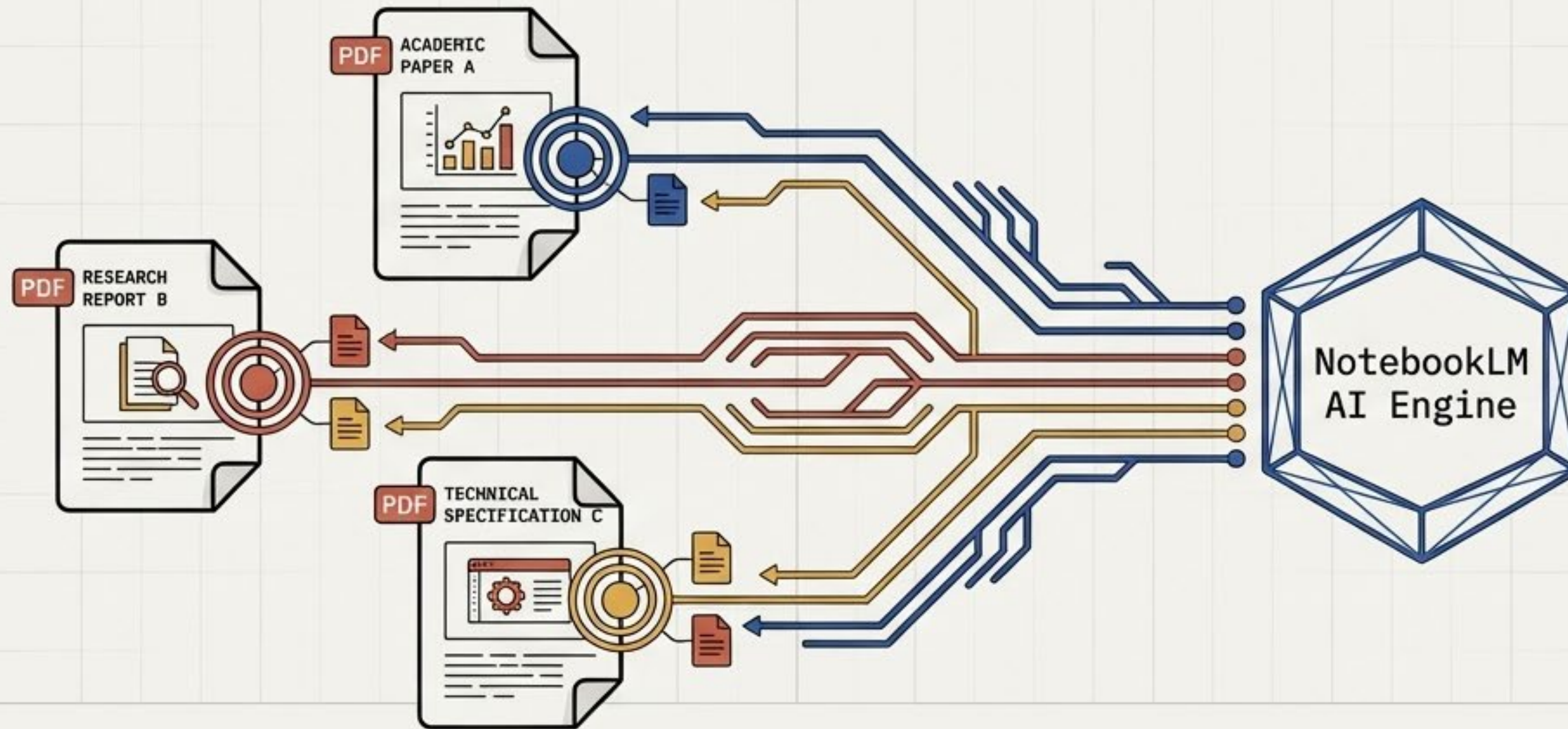


Use
Notebook
LM better
than 99%
of
students

<https://youtu.be/G7kDcCgLqFU?si=vv2qpGSX1cRVkxhK>

<https://notebooklm.google/>

[LIVE SANDBOX MISSION 02]: THE SYNTHESIS PROTOCOL



Execution Protocol:

1. Upload three entirely unrelated academic documents or PDFs.
2. Deploy a Chain-of-Thought prompt commanding the model to map logical intersections between the sources.
3. Evaluate the generated connections for academic validity and novelty.

Objective: Test the AI's ability to link disconnected information and accelerate the research phase.

[FINAL AUDIT]: THE EVOLUTION OF A PROMPT

Step A (The Baseline):

Write a raw 'Zero-shot prompt' asking AI to explain a complex topic from your taken modules. Log the output.



Step B (The Refinement):

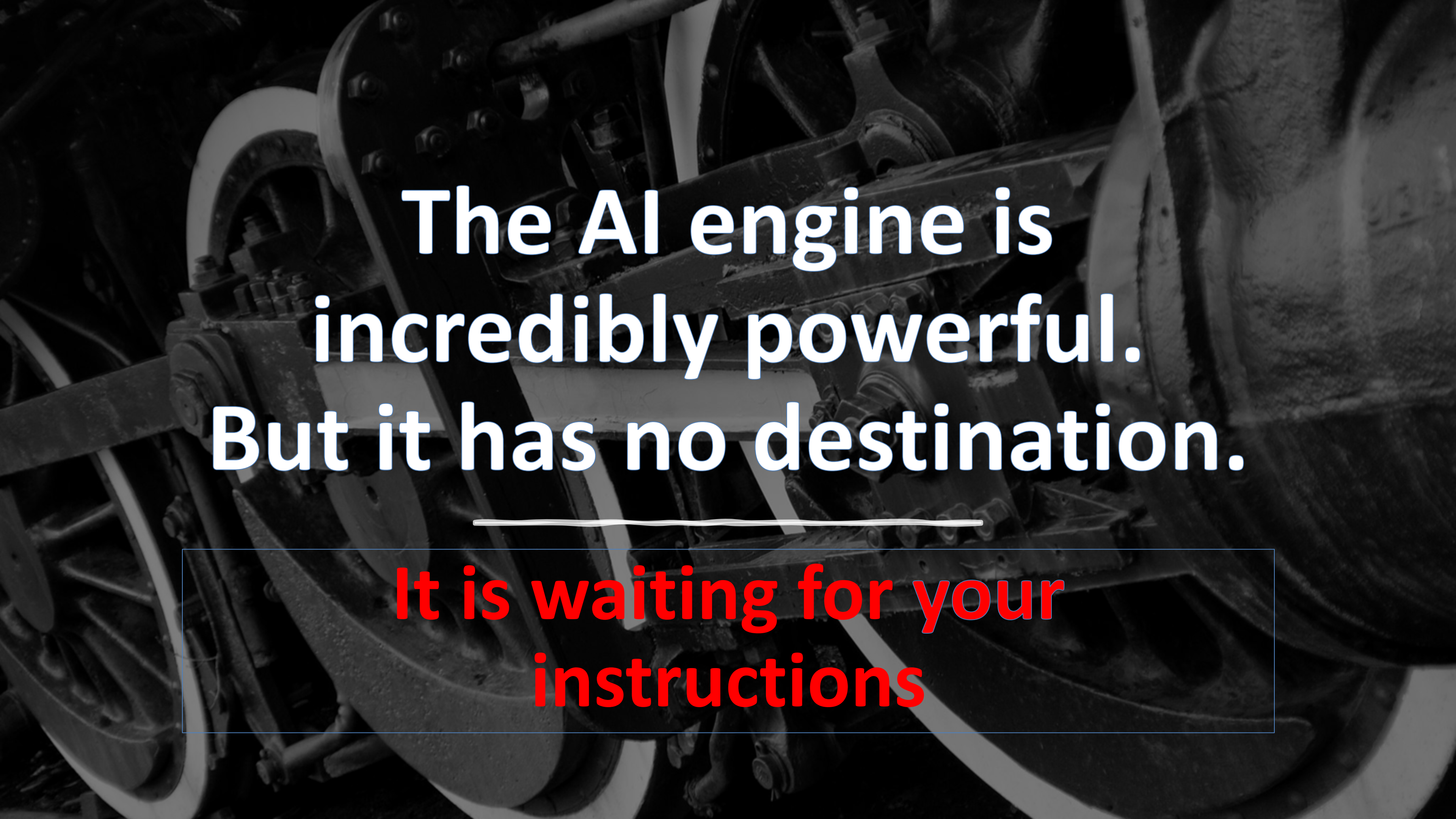
Inject the 4 C's. Apply 'Role-Based' or 'Chain of Thought' modifiers to tailor the exact same topic for a struggling first-year student. Log the new output.



Step C (The Critical Reflection - Human Override Required):

Submit both logs. Write an analysis pointing to specific sentences where your engineered constraints directly forced the model to change its behavior.





**The AI engine is
incredibly powerful.
But it has no destination.**

**It is waiting for your
instructions**